

How Bad Is It? Variation in Perceptions of Regional Economic Conditions

Amuitz Garmendia Madariaga^{a*}

^aDepartment of Social Sciences, Universidad Carlos III de Madrid, Getafe, Spain

*Corresponding author: Amuitz Garmendia Madariaga, Universidad Carlos III de Madrid, Calle Madrid 126, Office: 18.2.C.07, amuitz.garmendia@uc3m.es,
@amuitzgarmendia (orcid.org/0000-0003-2756-292X)

Amuitz Garmendia is an Assistant Professor of Political Science at Universidad Carlos III de Madrid and a Fellow Member of the Carlos III - Juan March Institute. She received her Ph.D. from Binghamton University, where she arrived as a Fulbright grantee. Before moving to Madrid, she was a Max Weber Postdoctoral Fellow at the European University Institute in Florence. She works on questions of federal political economy and in the theoretical and empirical implications for multilevel elections, political parties, and individuals' political behaviour.

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Drawing from recent theories of economic accountability that stress the contextual nature of attributed responsibility, this paper explores how variation in regional authority influences the interplay between objective regional economic conditions and perceived regional economic performance. The study uses a virtually unexplored set of surveys about public opinion in European Union regions in a broad sample of politically decentralized European democracies. Results demonstrate that while higher levels of self-rule increase the weight citizens place on economic outcomes when building their regional evaluations, higher levels of shared rule reduce it.

Keywords: decentralization; regional politics; European Union; economic evaluations; attribution of responsibilities

Even if crucial for democratic representation, only recently has scholarship started to explore the formation of policy evaluations in general, and economic evaluations in particular (Bartels 2002; Duch et al., 2000; Ebeid & Rodden, 2006; Evans & Andersen, 2006; Fortunato et al., 2018; Kayser & Peress, 2012; Stevenson and Duch, 2013). By and large, aggregate level models of economic voting have historically treated individuals as homogeneous in their reactions to their governments' economic performance (Mackuen, Erikson, & Stimson, 1992; Stigler, 1973). The demonstrated heterogeneity in the results of economic voting models questions the appropriateness of this conventional approach. The search of an explanation for the existing variation in individuals' aggregate level perceptions has motivated lately a deeper look at the nature of the economy that individuals can actually "observe". A scrutiny inspired by the fact that the "observable" economy is unlikely to be unique to all individuals, given it is constructed through "small, idiosyncratic, samples of mediated messages" (Stevenson & Duch, 2013, 307). This concern is interestingly addressed by Fortunato, Swift and Williams (2018, 467) when underscoring the necessity of paying more attention to "how differing distributions of economic messages may be aggregated by (or for) voters in an effort to ensure that scholars accurately model the mechanisms voters use to generate their perceptions of economic performance".

In fact, the idea of contextualizing economic information dates back to Powell and Whitten's (1993) seminal work, which showed that the degree of cohesiveness and visibility of governing institutions measured as a "clarity of responsibility index" significantly conditions the impact of economy on the electoral support for incumbent parties. The finding that economic voting is weakest in countries where multilevel governance is most prominent (Anderson, 2006) has recently put federal institutions under the spotlight of the scholarly debate and into the mentioned index (Hobolt et al.,

2013). Yet, even if we now know that vertical distribution of power contributes to blur responsibilities, we lack a proper understanding of the mechanism by which individuals build their perceptions about the economy in these contexts, and of the factors determining which economies are relevant in the process of aggregating information to evaluate economic performance across levels of government.

This paper contributes to fill this gap by analyzing the extent to which decentralization of power, and more specifically, variation in types and levels of regional economic decision-making authority influence the interplay between observed economy and perceived performance in these local jurisdictions. In a nutshell, I here argue that power distribution at the regional level (commonly categorized according to self and shared rule dimensions) acts as a significant facilitator or inhibitor in the process of motivating evaluations of observed regional economies, by intensifying or blurring the economic presence of governments at this level. Likewise, I argue that this institutional effect should be observable even after taking into account previously identified interfering contingencies on public opinion formation at the regional level, such as, the electoral support for regional parties in each constituency, or the level of territorial consolidation of a regime. I test this argument using a virtually unexplored set of Flash Eurobarometers (356, 427 & 472) on “Public Opinion in European Union Regions”, which were developed in an exhaustive hybrid combination of NUTS-I and NUTS-II units during and after a particularly “clear” tough time as the so-called Great Recession (Bermeo & Bartels, 2013) in Europe.¹ The availability of the outcome of interest (regional economy evaluations) in three different moments provides the necessary means and variation to explore the validity of the argument stated above.

The article proceeds as follows. The next section introduces a framework of reference regarding the literature on the intersection between subjective perceptions and

objective economic conditions. Afterwards, I delve into the theoretical argument by providing the means to understand the existence of systematic differences across territorial units when it comes to evaluating regional economies. Later, I present a detailed research design, and, subsequently, I proceed to comment the main results of the research. Interestingly, findings confirm the significant effect of authority distribution on the outcome of interest; that is, during and after the Great Recession, positive perceptions of regional economies tended to vary systematically depending on the actual decision-making power of these jurisdictions. Finally, the last section of the paper considers the main implications of the argument and analysis before concluding.

Subjective Perceptions of Objective Conditions

Scholars working on the relationship between economic performance and incumbent success have two kinds of available measures to account for the specific effect of existing conditions: subjective and objective ones. While subjective indicators include individual or aggregate level citizens' evaluations of the state of the economy and other further perceptions about the existence of observable economic changes, objective indicators represent aggregate statistics of economic activity. Notwithstanding the relevance of the extensive and systematic empirical work on economic voting, the use of these alternative "right-hand side" variables does not come without controversy. Praised by some as indicators of the state of the "real" economy (Van der Brug et al., 2007, 22), aggregate statistics have also been criticized for being "noisily-derived estimates of the observable economy, subject to political pressures and prone to subsequent revisions" (Fortunato, Swift, & Williams, 2018, 469; Stevenson & Duch, 2013) and seldom used given the methodological constraints they introduce when empirically modelling economic voting.

Actually, it is the search for variation at the individual level what motivates, to a large extent, the use of measures of subjective economic perceptions. A great deal of work in both political science and economics though, has asked the question of whether subjective economic assessments are accurate at either individual level or when averaged over a representative sample of individuals (Duch & Stevenson, 2008; Fraile & Pardos-Prado, 2014).

It is generally accepted that individuals form their opinions about the economy based on their views of past conditions (Fiorina, 1981).² These retrospective evaluations of the economy have been alternatively explained as a consequence of self-interested (“pocketbook”) judgments about personal economic conditions (Fiorina, 1981; Kramer, 1983), or as the outcome of respondents’ ‘sociotropic’ assessments of the economy (Kinder & Kiewiet, 1979). Although the latter behaviour contrasts sharply with almost all political economy models, which assume that voters are purely self-interested and care only about their own economic well-being (Meltzer & Richard, 1981), the corresponding findings have generally been substantiated with larger comparative datasets (Kinder & Kiewiet, 1981; Lewis-Beck, 1990). Nevertheless, beyond the assumption of perfectly informed instrumentally rational voters who reward incumbents with their vote when the economy is good and punish them when the economy is bad (Ferejohn, 1986), existing limitations and characteristics across different subgroups of a certain population do also affect the congruence between objective conditions and perceptions. Individual level differences on characteristics such as gender, race, age, socioeconomic status, political interest or media exposure contribute to the accuracy of individual level economic evaluations (Hibbs Jr, 1982; Krause, 1997; Zaller, 1992). Likewise, the latter are also influenced by partisan bias or elite level messages, shaping citizens’ perceptions and reactions to their environment (Bartels, 2002; Tufte, 1978). Yet, while biased and

inaccurate perceptions at the micro level have been considered to cancel each other out at the aggregate level (Page & Shapiro, 1992; Stimson et al., 1995), there are those who argue that these distortions endure in the aggregate (Kramer, 1983), since instead of being randomly generated noise, they are based on systematic individual level variation in, for instance, available information and media exposure, political attitudes, etc. (Duch et al., 2000). In fact, literature makes it clear that economic voting is both pervasive and variable, depending on the context. Responding to the inconsistent cross-national aggregate evidence, Powell and Whitten (1993) suggested that the impact of economic evaluations on vote choice might interact with the institutional context. Then, in political environments in which policy-making authority is widely shared, rational voters' ability to extract an informative signal about the competence of a party or a representative on economic matters declines (Duch & Stevenson, 2008). The study of economic perceptions in transitional regimes has also interestingly demonstrated that it takes time to observe the match between objective conditions and subjective evaluations in new democracies (Anderson & O'Connor, 2000). Other models situate perceptions in a particular spatial context, with authors like Kayser and Peress (2012) showing that national economic performance is contextualized relative to global performance, and that voters reward or punish incumbents for national deviations from global performance standards. Similarly, Fortunato et al. (2018) argue that voters incorporate economic information from other localities with similarly structured economies and in ways that are shaped by media messages. Moreover, public's reaction to an economic deterioration is usually larger than its reaction to an improvement (Parker-Stephen, 2013; Soroka, 2006), which is commonly known as a "grievance asymmetry" (Bloom & Price, 1975; Mueller, 1970; Stanig, 2013).

Then, regardless of whether it is useful to think of there being a single “real” or “true” state of the economy within some jurisdiction at a given time, literature has concluded that we certainly do not have a direct and accurate knowledge of this level. Instead, as it logically follows from the state of the art, the “relevant economy is the economy that voters can and do observe” (Stevenson & Duch, 2013, 307), even if this is not constant across them. Consequently, consistent with the literature on attitude formation and change (Zaller, 1992), I here argue that an individual’s perception of the economy reflects information differences and contextual considerations arising from the distribution of all the mediated messages about the economy in a certain jurisdiction for some interval of time (Fortunato et al., 2018; Stevenson & Duch, 2013). It is thus crucial to think in terms of this variation as a source of subjective heterogeneity when conducting aggregate level analyses at the regional level.

Decentralization of Authority and the “Observable” Regional Economies

Recent studies emphasize how the Great Recession posed historical coordination challenges to many of the comparative institutional examples of federal and quasi-federal countries in the Western world (Beramendi & Rogers, 2019; Kincaid et al., 2010). Ideal types of fiscal federalism models envision highly autonomous subnational governments and a clear separation of powers between layers of government (Oates, 1972) but these prototypical scenarios are far from the actual distribution of authority in the former systems (Rodden, 2006). Indeed, the existing complex and heterogeneous distributions of authority in federal and decentralized states makes responsibility attribution a daunting task for citizens (Arceneaux, 2006; Tuschhoff, 1999).

Based on the evidence on the effect of horizontal clarity of responsibilities on economic voting (Powell & Whitten, 1993), recent research has demonstrated that vertical distribution of authority undermines individuals' ability to administer political sanctions via evaluations of the economy (Anderson, 2006; Cutler, 2004; Hobolt et al., 2013; Wlezien & Soroka, 2010). Vertical distribution of authority alters the expected effects of economic outcomes on individuals' political judgements. A main mechanism explains such result: multiple levels of government impose additional information costs on individuals, influencing their ability to determine which level of government is responsible for economic and policy outcomes. There is proof of this relationship in contexts of global economic integration (Fernández-Albertos, 2006; Hellwig & Samuels 2007), supranational governance (Hobolt & Tilley 2013) and federal or quasi-federal structures. As an example of the latter, Anderson (2009) finds evidence of weaker economic voting in multilevel systems and shows that the transition to federal institutions in Belgium has resulted in an attenuated relationship between economic conditions and government support. León (2011) introduces a pertinent nuance to this research line by showing that the relationship between decentralization and clarity of responsibilities is contingent upon the balance of expenditure and fiscal powers across levels of government. According to the author, at low and high levels of decentralization there is always a level of government that predominates over the others making responsibility attribution easier, whereas at intermediate levels of decentralization, where levels of shared rule are high, this evaluation becomes problematic (León & Orriols, 2016). Consequently, according to this literature, subjective authority attribution is the underpinning mechanism motivating the way in which institutions shape any of the studied behavioral outcomes.

Evidently, the idea that individuals consider subnational economic conditions in any way in their evaluations is only plausible in countries where economic outcomes actually vary substantially across regions. As Ebeid and Rodden (2006) argue in their analysis of the effect of state level economic contexts in American gubernatorial elections, this can only happen in political systems where subnational governments enjoy certain level of decision-making autonomy. Then, notwithstanding the existence of national economic cues, which should ideally be equally available to citizens in all jurisdictions, the decentralization of decision-making authority points out to the potential existence of regional level outcome and (derived) perception variation. This is important given that decentralization of power in Europe is an extended and diverse practice. Indeed, the authority of regional jurisdictions in Europe varies more than that of national states; some of these are just central outposts for conveying and retrieving information while others exert more influence over the lives of people living under their rule than the central government itself (Hooghe et al., 2015). Besides, this heterogeneity pervades coountry borders with increasingly common examples of inter-regional formal power disparities: the existence of constitutional asymmetries ranges in Europe from simple mono-dimensional examples, as the Aaland Islands in Finland, to complex arrangements as those of the “*regioni a statuto ordinario e speciale*” in Italy. As it follows, if economic information has a contextual nature, citizens in the Basque Country do not sample from the same distribution of economic messages that citizens in Andalusia. This is because each of these Spanish regions has its own local economy, whose saliency to its citizens depends, largely, on policy-based performance, as well as on the type and depth of authority these regions hold.

Accordingly, the level of decision-making authority of a specific region, as well as the extent to which the latter is owned or shared with other levels of government,

should all be factors shaping how individuals build their perceptions about the state of their regional economies. As such, these perceptions might interestingly proxy individuals' subjective authority attribution to regional governments. Following Duch and Stevenson's (2008, 284-285) logic, a contextual theory of rational economic evaluation as the one presented here suggests that citizens are able to have an overall hint on the distribution of responsibility and that, along with other information about the competency signal, this information alters the weight they assign to performance in their evaluations of the economy. This weight is a function of the degree to which regional governments can be deemed responsible for subnational economic outcomes.

Then, in regions where authority is highly shared across levels of government, a blurred distribution of responsibilities should promote more acquiescent attitudes towards regional economies. A region with shared rule represents generally a jurisdiction whose fate is administratively, politically and fiscally connected to that of the central government through the institutionalization of intergovernmental platforms and mechanisms for vertical co-decision. As such, the responsibility of economic management is less clear and prevalent in citizens' judgments than in other jurisdictions, and consequently, given the diffusion of responsibility across levels, on average, these regions should have relatively more conformable economic evaluations.³ In contrast, the explicitness of responsibility distribution, as regional levels of self-rule increase, should decrease the average satisfaction with the state of the economy at the regional level. In autonomous jurisdictions, regional governments' decision-making is independent from central government's control, and thus the former enjoy a room for manoeuvring that lays the groundwork for a substantial capacity to influence policy outcomes. Hence, self-rule activates a mass of regional level critical observers and establishes the means for political

competition around them, which, on average, should reduce regional satisfaction with economic evaluations. Therefore,

- *Hypothesis 1.a: Regions with higher levels of shared rule authority have more positive aggregate level economic evaluations*
- *Hypothesis 1.b: Regions with higher levels of self-rule authority have more negative aggregate level economic evaluations*

As it logically follows from the previous argument, economic performance should be moderated by these average institutional effects. Specifically, it seems reasonable to expect that, in highly autonomous regions, economic outcomes might significantly increase or decrease the average satisfaction with regional economies. That is, where autonomy is high, economic outcomes should have the expected effects: GDP growth should improve public judgments while higher levels of unemployment should deteriorate them. Conversely, higher levels of shared rule should attenuate the direct link between outcomes and performance decreasing the relevance of the former on regional accountability processes via economic evaluations. That is, where decision-making authority is vertically intertwined, the effect of economic outcomes should be less prominent. Therefore,

- *Hypothesis 2.a: As the level of shared rule in a region increases, the effect of economic outcomes on aggregate level economic evaluations decreases*
- *Hypothesis 2.b: As the level of self-rule in a region increases, the effect of economic outcomes on aggregate level economic evaluations increases*

Research Design

Data

This paper relies on the novelty of up till now unexplored survey data on regional level perceptions in the European Union. The Flash Eurobarometers 356, 427 and 472 present a large scope view of public opinion on economy and politics in Europe at the regional level. The three were conducted using a hybrid combination of NUTS-I and NUTS-II regions.⁴ In total, 170 regions in 27 EU member countries were considered in 2012, while in 2015 and 2018 the number of regions raised up to 209 and 204 respectively in EU-28. This evident increase in the total number of regions included in the field work reveals the necessity of delving into the conceptualization of “regions” in this dataset by introducing some clarification (Lee and Rogers, 2019, 264).

First, even if by principle the NUTS system is designed to favour existing administrative units, the subdivisions at some levels do not necessarily correspond to governmental jurisdictions within a country. Then, these subdivisions are by no means standardized with respect to, for instance, a minimum level of decision-making authority. Therefore, while in 2012 NUTS-I regions in Germany corresponded to the sixteen actual Länder, NUTS-I in Portugal distinguished between the Continental units, as a single geographical group of *Comissões de Desenvolvimento Regional*, and the autonomous regions of Açores and Madeira. Second, even if they purposely share the same questionnaire, FB 356, 427 and 472 were developed based on different NUTS classification periods.⁵

Therefore, the models in the next section are estimated on evaluations that pertain to a specific type of regions in the three waves, that is, formally defined units with the

autonomy to elect their own governing bodies, whichever the amount of decision-making authority the latter hold. Regional level elections then work as a lowest common denominator assuring a sample of units that are territorially defined (with clear borders) and salient to their inhabitants. Following this description, I consider eleven countries (across different European Union latitudes and enlargement waves) in the end.⁶

(Table 1 here)

Dependent variable

The Flash Eurobarometers 356, 427 and 472 inquired a representative sample of citizens in the above-mentioned NUTS units about their perceptions on their regional economies. The question asked respondents whether the economy at that territorial level was perceived as very good, rather good, rather bad or very bad (*How would you judge the situation in each of the following? The situation of the Regional Economy*). I employ the aggregate percentage of respondents indicating they perceived their regional economies as performing goodly (very good + rather good).⁷ If, during a recession, the economy is likely to dominate any other issue concern (Bloom & Price, 1975; Lewis-Beck & Paldam, 2000; Soroka 2006), such an extensive map of the observable economy at the regional level in the middle of the Great Recession and after so in Europe presents an outstanding opportunity to understand the extent to which vertical distribution of decision-making authority affects the way evaluations of economy are formed and delivered. Put it differently, an economic situation that is obviously bad across a sample of countries of interest could actually lay the groundwork for the identification of behavioural deviations – from latent objective evaluations of the economy – due to systematic regional differences.

The data are also available at the individual level and the results of a non-reported three level ANOVA model show that all variance components are significant, suggesting that there is statistically meaningful variance in economic evaluations at all three potential levels of analysis (individual, regional and national). However, the lack of a comprehensive set of individual level socio-economic and political covariates relevantly limits this endeavour. These surveys fail to incorporate much of what literature identifies as explanations (information, political attitudes, personal experiences and socioeconomic characteristics) of formation and variation in economic perceptions (Duch, Palmer & Anderson, 2000). Therefore, without neglecting the importance of factors that directly shape the manner in which individuals draw their evaluations about economy, this paper will identify the institutional and political explanations for regional variation in aggregate subjective economic evaluations, once controlling for main regional macroeconomic indicators.

Independent variables

Regional level

The primary regional level explanatory variable is the Regional Authority Index (Hooghe et al., 2015), which measures the institutional power of individual regions, accounting for the sum of their shared and self-rule.⁸ *Self-rule* is the autonomous authority exercised by a regional government over those who live in that specific region and ranges from 0 to 18.⁹ *Shared rule* instead measures the authority exercised by a regional government or its representatives in the country as a whole by co-determining national legislation, policy outcomes or constitutional change and ranges from 0 to 12. Each of these domains of

regional governance can be further decomposed into dimensions of decentralization. Given the nature of the dependent variable, the focus here will be on regions' economic or fiscal authority. Particularly, I specify as economic self-rule the sum of *tier_fiscauto* (the extent to which a regional government can independently tax its population) and *tier_borrowauto* (the extent to which a regional government can borrow). For economic shared rule I consider the sum of *tier_fisccon* (the extent to which regional representatives co-determine the distribution of national tax revenues) and *tier_borrowcon* (the extent to which a regional government co-determines subnational and national borrowing constraints). Hence, these restricted domains capture regions' fiscal and borrowing self and shared rule, which range from 0 to 7 and 0 to 4 respectively. Since the Index measures, on the one hand, those powers that could lead to clarify responsibilities, and on the other hand, those that tend to blur them, both types of authority are introduced separately in the models (see Figure 1). These measures are available for a variety of administrative divisions by country, which facilitates the use of comparable measures across different levels of NUTS. Besides, the use of these variables at the regional level becomes especially relevant given the asymmetric distribution of territorial autonomy in some of the countries in the sample (for instance, Belgium, Italy, Spain, or the United Kingdom).

(Figure 1 here)

A set of economic indicators is also used as covariates. The strategy here is to regress economic perceptions on objective economic conditions in order to control for congruence, assuming that higher unemployment rates should consequently generate lower evaluations of the economy at the regional level, or vice versa. *Regional Unemployment* rate comes from the Eurostat's data on "Regions and Cities". In addition, *Regional GDP per capita* accounts for the relative wealth of each region (Eurostat).¹⁰

Finally, I control for any possible effect of the size of a region on the outcome of interest by introducing the natural logarithm of the total squared kilometres of a region's *area*.

Yet, partisanship and ideological orientation also shape how citizens observe their surrounding conditions. Building upon vast political science literature on retrospective economic voting and much wider social psychology literature on the attribution of responsibility, a recent research line has shown that partisanship is not only an important influencer of evaluations when predicting vote choices, but that it is an important predictor of who is thought responsible for performance outcomes in the first place (Malhotra & Kuo, 2008; Rudolph, 2003; Tilley & Hobolt 2011). This logic has also been extended to contexts of multilevel governance, as evidence shows individuals use partisan rationalizations when holding national and regional governments accountable (Brown, 2010; León et al., 2018; Toubeau & Wagner, 2018). Considering this, *Alignment*¹¹ is a dummy variable that takes the value of 1 when regional and national incumbents are members of the same state-wide party. The purpose here is to control for the existence of a vertical partisan connection and its potential contribution to clearing up the process of credit/blame attribution.

There are also territorially based group-serving biases that crosscut the classic left-right dimension. For instance, as Hobolt and Tilley (2013) have demonstrated, the support for Europe works as a systematic bias in the context of the European Union, with enthusiasts significantly differing from opponents when giving the EU more credit for positive outcomes in a variety of policy issues. Similarly, the existence of territorially concentrated minorities does not only provide a stable identity based electoral support for the corresponding representative ethno-regional political party or group, it provides a type of support that it is not subject to be extraordinarily changed depending on issues that could be located in a classic left-right dimension. These types of parties are relatively

immune against electoral punishment stemming from government performance, as conventionally defined in economic terms, given that their core dimension of competition is a territorial one (Elias et al., 2015). Since the salience of this dimension makes local autonomy a contested political issue across the board (Alonso, 2012), it seems reasonable to expect that the more prominent the territorial dimension in a region, the lower should be the effect of economic outcomes on regional economic evaluations (Pardos-Prado & Sagarzazu, 2019). That is, regionalism and its political/electoral saliency in a particular region should significantly boost the economic evaluations in the former.

Consequently, *Regional Parties' Vote* accounts for the percentage of votes of those organizations that compete and win votes only in one region, at the last regional level elections (see the Appendix). Then, even if regional parties might adopt a position on the classical left-right dimension (Alonso, 2012), I here understand regional parties as organizations that compete primarily in their core territorial dimension and that tend to focus their agendas on issues affecting only their regions, whereas state-wide parties focus their agendas on issues affecting groups throughout the country. Therefore, this variable serves as a proxy to understand the popular support the political defence of regional autonomy and territorialized interests enjoy in every unit of analysis. *Regional Incumbent* is a dummy variable that takes the value of 1 when a regional party holds the premiership of a regional government. Finally, to wrap it up, the *Effective Number of Parliamentary Parties* (ENPP, Laakso & Taagepera, 1979) captures the fragmentation of institutionalized competition in every region. ENPP is commonly used in the clarity of responsibilities literature as a measure of power diffusion that makes more difficult for voters to identify a clear alternative to incumbents and thus attenuates direct accountability processes via vote (Anderson, 2000). Yet, this voting regularity is both theoretically and empirically compatible with the fact that as the number of parliamentary

party alternatives to the government increases, the variance around average economic evaluations increases too (Lago-Peñas & Lago-Peñas, 2010). Therefore, ENPP aims at controlling for this potential source of variation in the dependent variable.

National level

This article builds on research on attitudes towards federalism and decentralization to explain the existing country-level variation in the outcome of interest. The logic here follows the observation that respondents in long-standing federations tend to agree on the fact that a federal form of government is preferable to any kind of government (half of all respondents in Mexico and more than two thirds in Canada and the U.S. (Kincaid & Cole, 2010). Given the contested multilevel incentives in a federation – i.e. the chances to centralize and decentralize, and the derived territorial division of people into different regional groups – the more consolidated a decentralized regime is, the higher should be the knowledge about its institutional structure. As it follows, it seems reasonable to expect that the duration of a particular territorial structure should be connected to its support from a large popular majority. *Time Since Decentralization* measures the sum of the years since the adoption of the political decentralization reform to the regions in a country, and as such, it is introduced to control for variation in regime duration in the sample. Whatever the level of political competition in a territorial dimension in a country, which is already captured by previously mentioned political variables at the regional level, *Time Since Decentralization* accounts for the variance related to regime consolidation.

The Model

Using hierarchical linear models allows for the simultaneous control of regional, national and yearly sources of variance in the dependent variable, accounting for the intra-class correlation of the errors by avoiding inflated Type I error rates. Figure 2 shows to what extent is the within-cluster dependence due to between-cluster heterogeneity in the dependent variable. With an intra-class correlation (ICC) of 0.45, 0.38 and 0.31 in 2012, 2015 and 2018 respectively (around 30-40% of the variance in regional economic evaluations in the whole sample is due to differences across countries), this graph illustrates the evident distance between the units of analysis (dots), their within-cluster means ($\bar{x}$'s), as well as the sample grand-mean by year (cross-cutting lines). As it can be observed, the 20-point improvement between first and last moments in the outcome of interest confirms the economic recovery that many of these regions have experienced in the last decade.

(Figure 2 here)

Although having small countries with low numbers of regions appears to be non-problematic for estimation, the number of countries or groups in the sample is in the margin to assure the unbiasedness of the estimates and avoid the problems associated with maximum likelihood inference for multilevel models when the number of groups is small (Stegmueller, 2013). Since the goal is to estimate basic varying intercept linear three-level (region, country, year) models under a restricted maximum likelihood (Elf et al., 2016), with a single country-level covariate and no further statistical restrictions, having an eleven country sample appears as a sufficient condition for the robustness of the proposed estimation.

Results

Table 2 presents the results of the relationship between the aforementioned economic regional authority measures and positive regional evaluations of the economy. Models 2 and 3 incorporate the rest of the covariates. There is a highly significant relationship between economic self-rule and shared rule, and the dependent variable across models. Indeed, as specified in Hypotheses 1, while, *ceteris paribus*, regions with higher levels of fiscal and borrowing autonomy tend to experience more negative economic evaluations, this effect seems to be neutralized as the shared rule of a region in these issues increases. Interestingly, these correlations are quite robust to alternative model specifications including potential confounders, which all show strong statistically significant effects following the conventional expectations.

In this regard, it is worth underscoring the behaviour of the political indicators in Models 2 and 3. For instance, as the fragmentation of party systems increase, and thus, regional political competition becomes more contentious, economic evaluations suffer a considerable decreasing tendency. The opposite seems to happen with *Alignment*, which consistently boosts positive economic perceptions. The mechanism behind this result is commonly justified as the positive effect of having a political link with the central government incumbent, which as the literature demonstrates, it usually entails tangible and intangible externalities (León et al., 2018). With respect to the electoral presence of regional parties, there are some interesting regularities in the results: the effect of having a regionalist premiership is overwhelmingly large and positive on evaluations, while regional parties' vote percentage has an average negative effect. Results show how, once the variance arising from *Regional Incumbency* is controlled for, the effect of *Regional Parties' Vote* deteriorates aggregate economic evaluations. There might be two groups of regions here: those where regionalist attitudes are dominant electorally and politically, and thus higher vote percentages for regional parties might show a competition with other

regional parties in the opposition, and those where regional parties are an electoral minority and the vote they receive might be a way of externalizing dissatisfaction with the *status quo*.

(Table 2 here)

Table 3 presents the tests for the stated interactive effect – in Hypotheses 2 – between regional authority dimensions and unemployment, a readily accessible economic cue. The results confirm the enforcing or attenuating – depending on the dimension – effect of regional institutional contexts on aggregate evaluations, which are accordingly graphed in Figure 3 for a more detailed analysis. As observable, the average marginal effect of unemployment, which is around 10% in the sample, and in the model is calculated for each observation in the data and then averaged, decreases significantly positive economic evaluations when a region surpasses a level of economic self-rule of around 2. After this point, the effect of unemployment is significant and negative, confirming that once local authority becomes salient to the inhabitants of a jurisdiction, the relevance of specific policy outcomes in accountability processes increases too. Conversely, as the levels of shared rule and co-decision in a region increase, the effect of the same average level of unemployment on evaluations decreases, somehow tempering the previous effect.

(Table 3 here)

It is important to interpret these results with some perspective and nuance. First, as shown in Figure 2, most of the countries in the sample have, on average, a mix of both types of regional authorities. Indeed, some of the regions with considerable levels of self-rule, also tend to enjoy moderate levels of shared rule. Second, as hypothesized, two co-occurring conditions¹² can have opposite effects. Yet, the argument here is not that shared rule reverses the effect of unemployment, working as a sort of redemption mechanism for

regional governments. Instead, the idea here is that as institutional frameworks reduce the ability of citizens to understand “who is responsible for what” (León, 2011), the prevalence of economic outcomes in regional evaluations decreases, attenuating the average effect of economic indicators on subjective perceptions. Accordingly, certain particular combinations of self and shared rule might cancel each other’s effect out, reducing nearly to zero the result of regional economic outcomes on regional level economic evaluations.¹³

(Figure 3 here)

To better recognize the effects in Hypotheses 2, Model 6 reports the results of a regression with both interactions run together. Figure 4 shows the overlaid average marginal effects of economic self and shared rule on evaluations, as the unemployment rate varies. The increasingly separating, and statistically significant, paths of both dimensions interestingly illustrate the argument in its full scope.

(Figure 4 here)

Concluding Remarks

This paper has demonstrated that at a time of clear and unquestionable economic depression, other macroeconomic indicators and socio-political controls being equal, there are systematic differences regarding how citizens evaluate their regional economies in decentralized democracies in Europe. This is true even under perceived symptoms of economic recovery. By focusing on recent theories of economic accountability that stress the contextual nature of responsibility (Ebeid & Rodden, 2006; Kayser & Peress 2012, Fortunato et al., 2018), this article has presented the first comparative test of how regional

level institutional design variation influences how citizens in these regions perceive the “observable” economy.

Interestingly, results demonstrate that the type of distribution of responsibility across levels, along with other information about the competency signal, alters the weight that citizens assign to performance in their evaluations. Throughout the world, vertical distribution of authority is becoming more convoluted and the institutionalization of asymmetric autonomy is diversifying the levels of responsibility that regions exert. Accordingly, the way in which this institutional complexity intervenes in processes of individuals’ public opinion formation is becoming one of the most relevant challenges to multilevel governance.

Moreover, this paper has also contributed to uncovering pro-periphery bias in these political systems. Findings show that the electoral presence and strength of regional parties positively correlates with a more benign evaluation of regional economies. Results thus confirm the necessity of taking seriously the presence of a territorial second dimension of party competition in many European countries. Similarly, as the scarce literature on attitudes towards federalism would suggest, the temporal consolidation of the territorial regime for the countries in the sample has also shown a slightly significant positive effect on regional economic evaluations. While the latter outcome is modest and is only captured indirectly, it sheds light on the idea pervading this research endeavour: we must acknowledge the multilevel structural sources of subjective heterogeneity.

Future research might examine the compatibility and tensions between the economic and political geography conditions that shape this proven aggregate behaviour, as well as a third level of variation accounting for individuals’ information levels, preferences and incentives. More relevantly, future research should evaluate the extent to which the findings of this paper contribute to better explaining the existing patterns of

economic voting at the regional level. Recent efforts in the literature underscore the relevance of deepening our understanding of regional electoral dynamics (Toubeau & Wagner, 2018) by overcoming what is commonly identified as the pitfalls of “methodological nationalism” (Schakel & Jeffery, 2013). In this regard, the opposite effects of self and shared rule represent an important finding with crucial implications for the future use of the Regional Authority Index (Hooghe et al. 2016) in regional accountability models. In brief, this paper sheds light on why some regional incumbents might win elections even in the presence of bad economic performance.

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¹ Around 50% of the countries in the sample were in recession at the time of the field work of the FB 356, that is, the third quarter of 2012 (OECD-Quarterly National Accounts).

² Other works have demonstrated that rather than being purely myopic, the public can also be foresighted, incorporating into its' judgment more sophisticated expectations about the prospects of the economy (Alesina and Rosenthal 1989, MacKuen et al. 1992, Norpoth 1996).

³ The direction of the effect of the lack of clarity of responsibility is consistent with Powell and Whitten (1993, 402) when stating "*Table 2 shows the effect of the measures of political conditions on incumbents' vote gain or loss (...) As each condition is expected to blur political accountability, we expect each of the coefficients for the responsibility conditions to be positive*".

⁴ NUTS (Nomenclature de Unites Territoriales Statistiques) is a hierarchical system developed by EUROSTAT for dividing the economic territory of the EU for several statistical and policy analyses. NUTS-I (major socio-economic regions > 3000000 inhabitants); NUTS-II (basic regions for the application of regional policies > 800000 inhabitants); NUTS-III (small regions for specific diagnoses > 150000).

⁵ The NUTS 2010 classification was valid from 2012 to 2015. The NUTS 2013 classification was valid from 2015 to 2018. The current 2018 classification is valid from 1st January 2018 and lists 104 regions at NUTS 1, 281 regions at NUTS 2 and 1,348 regions at NUTS 3 levels.

⁶ Given that twelve countries changed their NUTS classification from 2012 to 2015, these are the eleven decentralized countries out of those seventeen that maintained a consistent NUTS level across waves. With respect to the territorial reform suffered by France in

2016, in 2018 only unchanged regions remain in the sample, unbalancing the number of observations in this country.

⁷According to the technical specifications of these Flash Eurobarometers “*The basic sample design applied in all countries is multi-stage random (probability). TNS have developed their own RDD sample generation capabilities based on using contact telephone numbers from responders to random probability or random location face to face surveys, such as Eurobarometer, as seed numbers. The seed numbers are stratified by NUTS2 region and urbanisation to approximate a geographically representative sample.*” Moreover, as detailed in the Appendix, the question used in this article (q1_2) has, at least, a 90% response rate in every region in each wave.

⁸ The latest available score of the RAI is the one for 2010 as presented by Hooghe et al. 2016. Given the time frame of this project, I here employ Schakel’s (2018) updated scores.

⁹ In the Appendix, I run the same models using the full measures of self and shared rule. Although, logically, the magnitude of the hypothesized effects declines, the results retain a strong significance, reinforcing the contribution of this paper. The effects hold even with a broader objective measure of authority, which underscores the importance of considering context based perceived authority when understanding regional level outcome evaluations.

¹⁰ Given the dates of the field research in each wave, both variables use the data from t_{-1} .

¹¹ *Alignment*, *Regional Parties’ Vote Percentage*, *Regional Incumbents* and the *Effective Number of Parliamentary Parties* at the regional level are all variables constructed based on publicly available data on multilevel elections in the studied countries’ governmental and national statistics institutions’ online sites.

¹² Economic self and shared rule's correlation coefficient is 0.48, while the correlation of the full measures of self and shared rule is 0.63.

¹³ The last section of the Appendix is devoted to further analyze the substantive implications of these results.